

Project Design Phase Proposed Solution Template

Date	01 Oct 2026
Team ID	SWTID-2026-4457
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	In ServiceNow, standard role-based ACLs cannot decide record visibility based on the data inside a record. Sensitive records (for example, student details in the Institution Details table) can therefore be seen by users who should not see them, such as users from other branches (ECE, CSE) viewing EEE branch records. The problem is to restrict record access so that a user can view, create, edit or delete a record only when both their role and the record's field value (Branch) meet defined conditions, while administrators retain full access.
2.	Idea / Solution description	A Script-Controlled ACL uses a custom script to check specific field values and user conditions before granting access. A user can access a record only when the ACL script evaluates to true. • Custom table u_institution_details stores student records with fields such as Student Roll Number, Student Name, Faculty Name, Branch (ECE, EEE, CSE), Email, Phone Number and Description. • Custom roles bb1, bb2, bb3, bb4 are created and assigned to a test user (EEE User). • Read ACL: Advanced ACL with data condition <i>Branch is EEE</i> and a script that returns true for admin or bb1 and false for everyone else. • Create ACL (role bb2), Write ACL (role bb3) and Delete ACL (role bb4) provide controlled access to each operation. • Verification: by impersonation, bb1 users see only EEE records, users without the role see no records, and admins see all records.

S.No.	Parameter	Description
3.	Novelty / Uniqueness	- Combines role checks with field-value (data condition) checks in a single script-driven decision. - Gives record-level security instead of only table-level or role-level security. - Separate ACLs for read, create, write and delete allow least-privilege, operation-wise access. - Admin override keeps full access for administrators. - Flexible: the script can be extended to other fields, groups or business rules without changing the table.
4.	Social Impact / Customer Satisfaction	- Protects sensitive personal data (names, emails, phone numbers) from unauthorized viewing, modification, creation and deletion. - Builds trust among students, faculty and administrators that their data is handled securely. - Supports data privacy and compliance expectations in institutions. - Users see only the records relevant to them, giving a cleaner and simpler experience. - Reduces accidental or malicious data changes, improving reliability.
5.	Business Model (Revenue Model)	- Offered as a reusable ServiceNow security component or accelerator that organizations can install in their instance. - Revenue from implementation and customization services for educational institutions, healthcare, finance and other organizations. - Subscription-based support and maintenance, including periodic security reviews and ACL audits. - Training and certification workshops on ACL design and record-level security.
6.	Scalability of the Solution	- Can be applied to any table by changing the table name, field and role conditions. - Supports more branches, departments, roles and large record volumes, since ACLs are evaluated natively by the platform. - Works consistently across forms, lists and Service Portal interfaces. - Can be packaged in an update set or scoped application and moved across instances (Dev, Test, Prod). - Can be extended with group-based or script-include logic for complex enterprise security rules.